

Curiosity-driven learning for visual control of nonholonomic mobile robots

Takieddine Soualhi*, Nathan Crombez, Alexandre Lombard, Stéphane Galland, Yassine Ruichek

Abstract—In this paper, we study the problem of visual servoing of nonholonomic mobile robots. Achieving precise positioning becomes particularly challenging within the classical approaches of visual servoing, primarily due to motion and field-of-view constraints. Previous work has demonstrated the effectiveness of deep reinforcement learning in addressing visual servoing tasks for robotic manipulators. In light of this, we propose a novel deep reinforcement learning framework that integrates deep recurrent policies and curiosity-driven learning to tackle the problem of visual servoing of nonholonomic mobile robots. First, we analyze the influence of the nonholonomic constraints on control policy learning, and subsequently, we evaluate our approach on both simulated and real-world environments. Our results demonstrate the superiority of our model in terms of spatial trajectories and convergence accuracy compared to the existing approaches.

I. INTRODUCTION

A. Motivation

In robotics, the problem of visual control of nonholonomic mobile robots is a long-studied challenge [1]. Visual servoing (VS) refers to a set of methods that utilize computer vision data to control the motion of dynamic systems, such as robots [2]. VS paradigm can be formulated as follows: given a goal defined by an image taken at a desired pose, the robot must accurately calculate linear and angular velocities to reach this desired pose. The primary objective of VS is to minimize an error between visual features extracted from a desired image and images acquired by the robot during the process.

The main challenge with the VS of nonholonomic robots is the limitations imposed by motion constraints. These constraints can make it difficult, and even impossible, to reach specific configurations without losing the desired visual features from the camera's field-of-view (FOV)(Fig. 1).

Several methodologies have been proposed to tackle the VS of nonholonomic mobile robots. The approach in [3] introduces a feedback control system to ensure that the desired visual content is always kept in the camera's FOV. The authors in [4] propose a novel motion estimation technique that leverages the continuity property of robot movements to enhance robustness against FOV loss problem. The work presented in [5] suggests adding a pan and tilt unit onboard to increase the camera's degrees of freedom (DOF). The paper [6] suggests using omnidirectional cameras to guarantee the visibility of desired visual features. Similarly, the authors in

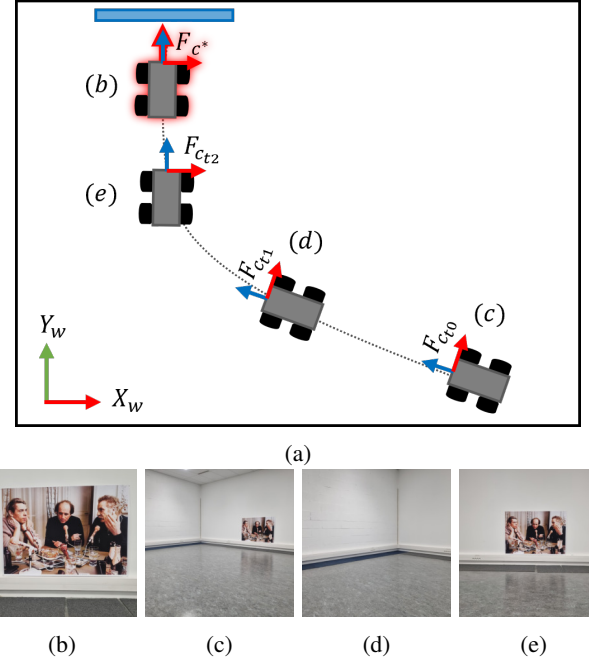


Fig. 1: Scenario where traditional VS methods fail due to visibility/motion constraints: (a) The trajectory that the robot has to follow to reach the desired pose, (b) The desired image captured at the desired pose, (c), (d), (e) represent the robot's views in poses (c), (d), and (e) respectively.

[7] extend the application of omnidirectional vision to the photometric VS of nonholonomic mobile robots.

In parallel, deep learning methods have achieved remarkable success over the last decade in computer vision tasks [8]. These advances have contributed significantly to breakthroughs in applications of deep reinforcement learning (DRL) to real-world problems, particularly in vision-based robotics. Although the problem of learning VS for robotic manipulators has been studied in previous works [9], data-driven VS of nonholonomic mobile robots remains largely unexplored. Thus, in this work, we investigate whether DRL is capable of learning a control policy that can make a nonholonomic robot precisely converge to a desired pose.

B. Related work

This section provides an overview of the related work, focusing on three key areas: target-driven navigation, learning VS, and intrinsic curiosity in DRL. The first two areas share the objective of controlling a robotic system using desired and current images as input. Target-driven navigation, unlike

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VS, uses DRL to guide a mobile robot towards a visual goal without considerations for precision at convergence.

Several works have been proposed to learn VS using DRL approaches. Authors in [10] presented a recurrent convolutional neural network (CNN) model to tackle the problem of VS of 6 DOF systems. The same authors extended the previous approach by using Monte Carlo estimation with recurrent policies, showcasing improved results [9]. Joshi et al. [11] used images from a multi-view camera setup as input to learn a VS control policy, demonstrating that the multi-view approach increases the accuracy of the VS task compared to classic single-view approaches. DRL approaches typically train in simulations due to safety and sample efficiency requirements, which pose challenges when deploying on real robots. This issue was addressed in [12], which utilized a generative adversarial network to translate real images into the synthetic training data domain. However, these methods are tailored for 6 DOF robotic manipulators.

Closely related to the VS of nonholonomic mobile robots is target-driven navigation. Authors in [13] were the first to propose a deep siamese network and an asynchronous actor-critic method to introduce the problem. The paper [14] showed that incorporating deep recurrent policies and auxiliary tasks benefits the navigation task. The paper [15] extended previous works by augmenting their model with an episodic memory, achieving state-of-the-art results on the Gibson dataset [16]. Despite the progress made in addressing this problem, current methods still struggle when the desired image is far from the initial pose and exhibit limited generalization capabilities. The works most closely related to ours are [17] and [18], both addressing the challenge of FOV loss and learning VS of nonholonomic mobile robots. In [18], the authors utilized dense image correspondence as input features to train a deep Q-network (DQN) with a novel-shaped reward function. In contrast, [17] proposed incorporating keypoint features from both desired and current images as additional inputs for learning a control policy. However, the approach in [18] requires training the dense correspondence feature extractor separately. Both methods assume the robot's convergence within a large distance from the desired pose. Additionally, the robot is trained to use constant control velocities, potentially resulting in unstable behavior at convergence. Moreover, their training is conducted in simulation without transferring the learned policies to real robots.

Exploration is the process by which an agent takes actions to discover new states in the environment, enabling learning about dynamics, transition probabilities, and action rewards. However, overexploration can lead to suboptimal actions with lower immediate rewards, while underexploration may cause premature convergence to suboptimal policies. Recently, intrinsic motivation [19] has emerged as a promising approach to enhance exploration in DRL. Intrinsic motivation encourages agents to explore the environment by providing additional rewards based on the novelty or complexity of the states encountered. There are several approaches for modeling intrinsic motivation in RL agents. Count-based methods

assign exploration bonuses based on how many times a state has been visited. The bonus pushes the agent towards searching for rarely visited states rather than common ones [20]. Another line of research is prediction-based exploration methods where the agent is rewarded for improving its representation using a predictive model to estimate the agent's familiarity with the dynamics of the environment [21]. These methods come with many disadvantages, such as the need for function approximation and knowledge fading. Episodic exploration methods were proposed to enhance the agent's exploration by either maintaining an episodic memory [22] or evaluating a statistical distance between old and current policies [23].

C. Contributions

We propose leveraging curiosity-driven learning [19] to train a nonholonomic mobile robot to solve a VS task. Curiosity-driven learning involves defining an exploration bonus to reward the DRL agent for visiting novel states. We show that motion constraints make it difficult for a nonholonomic robot to efficiently explore the environment and, consequently, learn complex trajectories to reach the desired pose. We evaluate our method in a simulated environment and validate its transferability and usability on a real robot, demonstrating its effectiveness in addressing VS tasks.

Our contributions can be summarized as follows:

- A novel DRL model that combines deep recurrent policies and exploration with intrinsic rewards using Rényi divergence.
- Comparisons with traditional image-based visual serving (IBVS) and the existing DRL approach [18], along with evaluations of the advantages and limitations of using DRL for VS of nonholonomic mobile robots.

D. Outline

The remainder of this paper is organized as follows. Sec. II describes the formulation of the VS problem within the context of DRL. Subsequently, Sec. III presents our proposed DRL approach. Sec. IV describes the experimental setup, training details and the compared methods. Sec. V showcases simulation results, comparisons, and real experiments. Lastly, Sec. VI presents conclusions, limitations, and future work.

II. DRL FOR THE VS OF NONHOLONOMIC MOBILE ROBOTS

In this section, we present a DRL formalization of the VS of nonholonomic mobile robots. This involves articulating a systematic framework to address the VS task using DRL approaches.

A. Problem statement

The goal of this paper is to study learning VS for a nonholonomic mobile robot with only two controllable decoupled velocities. Consider a scenario in which a mobile robot operates, aiming to reach a desired pose defined by an image. Most standard VS methods are based on the computation of an interaction matrix that represents the relationship between the camera displacements, and the motion

of the visual features. This Jacobian is then used to compute a feedback control law. The objective of a VS task is to reach a desired pose, denoted as $\mathbf{p}^* = (x^*, y^*, \beta^*)$ with (x^*, y^*) representing the desired position, and β^* is the desired orientation. This objective is achieved by minimizing an error between visual features extracted from the current and desired images captured at $\mathbf{p}_t = (x_t, y_t, \beta_t)$ and $\mathbf{p}^*$ respectively, where (x_t, y_t) represents the robot's position at instant t , and β_t represents its orientation. DRL methods constitute a powerful paradigm for learning a representation that directly maps the desired and current images to control velocities, without relying on any hand-crafted features. Similar to classical VS methods, our proposed DRL method relies solely on images as inputs. Spatial poses are used only to compute rewards during training.

B. Problem formulation

We propose learning a goal-conditioned stochastic policy function π through DRL, using a deep neural network as a nonlinear function approximator. Given only RGB images as input, we assume that VS tasks can be viewed as Partially Observable Markov Decision Processes (POMDPs), since the images do not explicitly contain information about spatial pose and velocity [24]. The POMDP is defined by (S, A, O, P, R) as follows: The state space S consists of the spatial pose of the robot, which is not accessible. The action space, observation space, and reward set are denoted by A , O , and R respectively, and will be detailed in the following sections. $P(s' | s, a)$ is the set of conditional transition probabilities of $s' \in S$ for the execution of action $a \in A$ in state $s \in S$.

The robot is initialized in a state s_0 , where it learns the optimal sequence of actions (control velocities) to visually align the currently acquired image with the desired one. At each time step t , the robot perceives an observation o_t , takes an action $\mathbf{a}_t$, and receives a reward $r_t(\mathbf{a}_t, o_t)$. The action $\mathbf{a}_t$ is sampled as follows:

$$\mathbf{a}_t \sim \pi(o_t | \theta) \quad (1)$$

where θ denotes the parameters of the policy neural network.

C. Learning setup

1) *Observation space*: Observations consist of two RGB images $o_t = \{\mathbf{I}_t; \mathbf{I}^*\}$ where $\mathbf{I}_t$ and $\mathbf{I}^*$ are the current image, that is, the image acquired at time t , and the desired image, respectively. The current and desired images are captured at $\mathbf{p}_t$ and $\mathbf{p}^*$ respectively using the robot's mounted RGB camera.

2) *Action space*: Real-world robotics applications involve the management of low-level mechanics and complex dynamics. This requires the use of a physical engine to handle the underlying physics of such low-level control and transferability to the real world. Additionally, precise positioning tasks require high control sensitivity. Therefore, we train our agent in a physical simulator on continuous control of decoupled linear and angular velocities v_x and v_β . At time step t , the robot executes $\mathbf{a}_t = (v_{x_t}, v_{\beta_t})$.

3) *Reward function*: At each instant t , the agent receives a reward $r_t = r_t^e + \eta r_t^i$, which is the sum of two rewards: an intrinsic reward r_t^i generated by the agent itself through self-supervision, and an extrinsic reward r_t^e related to the task. Here, η is a variable weighting parameter for the relative importance of intrinsic rewards. In this section, we focus on extrinsic rewards. The generation of intrinsic rewards is detailed in Sec. III-C. Our task reward function is defined as the Euclidean distance between the current and desired poses of the robot:

$$r_t^e = \lambda_d \cdot \exp\left(-K\sqrt{(x_t - x^*)^2 + (y_t - y^*)^2}\right) + \lambda_\beta \cdot \exp\left(-K\sqrt{(\beta_t - \beta^*)^2}\right) \quad (2)$$

where $\exp(x) = e^x$ is used to map the reward values between 0 and 1 to provide a standardized input for the neural network. λ_d and λ_β are weighting coefficients indicating the relative importance of reward components. K is a parameter to control the sparsity of the reward.

III. CURIOSITY DRIVEN LEARNING FOR THE VS OF NONHOLONOMIC MOBILE ROBOTS

This section introduces the proposed framework, which includes a policy model and an intrinsic curiosity model.

A. Visual domain randomization

To facilitate the transfer of our control policy to the real world, we apply random transformations on input images using classic data augmentation techniques. Specifically, we apply random changes in the image's brightness, contrast, saturation, and hue levels (Fig. 2). This helps the model better handle changes in lighting, backgrounds, and other environmental factors. Additionally, we apply two types of Gaussian noise: uncorrelated noise, which is sampled at the beginning of each training episode, and correlated noise, sampled at each time step t . These noise additions aim to replicate the noise generated by real cameras.



Fig. 2: Illustration of visual domain randomization applied to the input images.

B. Policy model

Due to the partial observability of the VS task, which necessitates an understanding of the relative spatial poses between the current and desired images, we propose using a deep recurrent actor-critic network. This framework has been shown to efficiently address the partial observability in DRL problems [24]. Fig. 3a illustrates our policy model

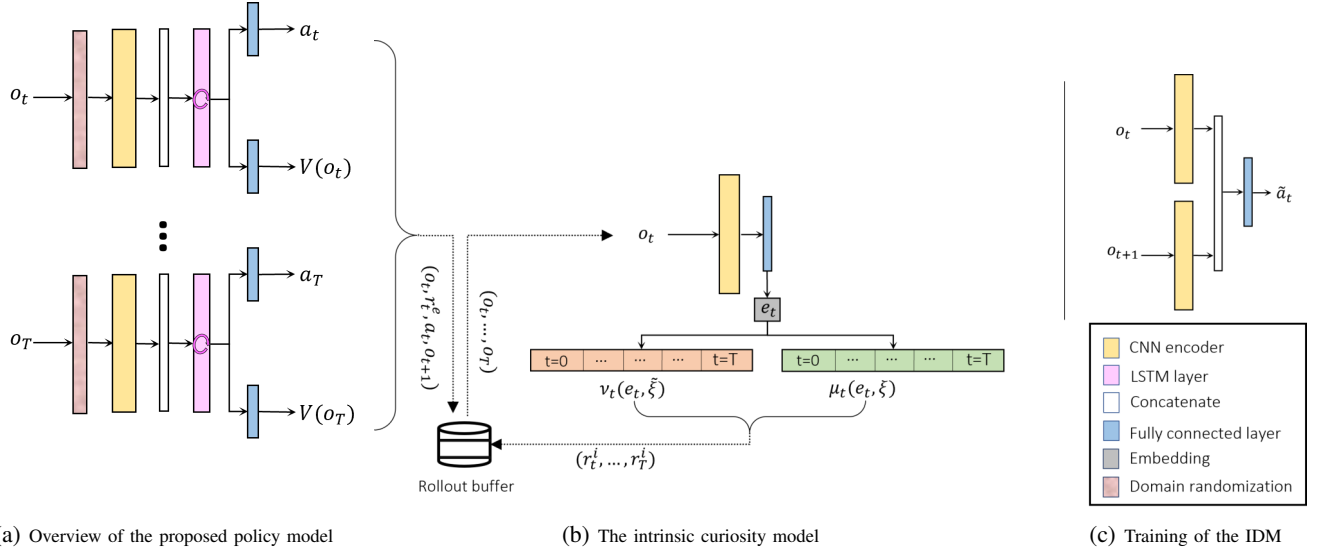


Fig. 3: (a) The base agent is a recurrent policy (RPPO) trained with a proximal policy optimization (PPO) loss [25]. During an episode $[t_0, \dots, T]$, observations, rewards, and actions are sampled and stored in a rollout buffer, (b) The inverse dynamics model (IDM) generates features from old and current policies observations. These features are then used to calculate intrinsic rewards by approximating the Rényi divergence, as shown in Sec. III-C. Once the intrinsic rewards are calculated, the rewards in the buffer are updated for each observation, (c) The encoder for the intrinsic curiosity model is an IDM trained to predict actions $\hat{a}_t$ given a pair of consecutive observations (o_t, o_{t+1}) using ground-truth actions from the rollout buffer.

to learn the VS task. Firstly, visual domain randomization is applied to the input images and then fed through a CNN encoder to extract visual features. These features are concatenated to create a unified joint representation of the observation. Subsequently, this joint representation passes through a long-short-term memory (LSTM) layer [26] to learn a temporal representation between observations. Finally, the model yields the robot control velocities $\mathbf{a}_t = (v_{x_t}, v_{\beta_t})$ and an estimate of the value function $V(o_t)$. The policy is trained to maximize the expected discounted return: $R_t = \mathbb{E}_\pi \left[\sum_{k=0}^T \gamma^k r_{t+k+1} \right]$, with $\gamma \in [0, 1)$ indicating the discount factor. T is the length of the episode.

C. Intrinsic curiosity model

The nonholonomic constraints make it challenging for the DRL agent to explore the environment. To address this challenge, we introduce an intrinsic curiosity bonus based on the approximation of the Rényi divergence [23]. The choice of this method stems from its consideration of both past and current policies when computing intrinsic rewards. It is important to note that the intrinsic curiosity model is used only during training to generate intrinsic rewards (Fig. 3b). During evaluation (i.e., deployment), only the policy model is used. Let $p_\pi(s)$ denote the state distribution induced by the current policy, $p_{\bar{\pi}}(s)$ represent the state distribution induced by the previous policy, and let α be the order of the divergence where $\alpha \in [0, \infty[$ and $\alpha \neq 1$. The Rényi divergence for policy π is defined as:

$$D_\alpha(p_\pi(s) \parallel p_{\bar{\pi}}(s)) = \frac{1}{\alpha - 1} \cdot \log \int_\xi p_\pi^\alpha(s) p_{\bar{\pi}}^{1-\alpha}(s) ds \quad (3)$$

This can be approximated using the following K-nearest neighbor (k -NN) estimator. Let $\xi = \{s_t\}_{t=1}^T$ and $\tilde{\xi} = \{\tilde{s}_t\}_{t=1}^T$ be the sets of visited states from two consecutive episodes. The Rényi divergence can be approximated as follows:

$$\begin{aligned} \hat{D}_\alpha^{k,T}(p_\pi(s) \parallel p_{\bar{\pi}}(s)) &= \\ \frac{1}{\alpha - 1} \log \left(\frac{1}{T} \sum_{t=1}^T \left[\frac{(T-1)\mu_k(\mathbf{e}_t, \xi)}{T v_k(\mathbf{e}_t, \tilde{\xi})} \right]^{1-\alpha} C_{k,\alpha} \right) & \quad (4) \\ \propto \frac{1}{T} \sum_{t=1}^T \left[\frac{\mu_k(\mathbf{e}_t, \xi)}{v_k(\mathbf{e}_t, \tilde{\xi})} \right]^{1-\alpha} \end{aligned}$$

where: μ_k is the Euclidean distance between $\mathbf{e}_t$ and its k -th nearest neighbor in ξ . v_k is the Euclidean distance between $\mathbf{e}_t$ and its nearest neighbor k in $\tilde{\xi}$. $\mathbf{e}_t = f_\phi(o_t)$ is an embedding vector produced by an IDM $f_\phi(o_t, o_{t+1}) = \hat{\mathbf{a}}_t$. $C_{k,\alpha} = \frac{\Gamma(k)^2}{\Gamma(k-\alpha+1)\Gamma(k+\alpha-1)}$, $k > |\alpha - 1|$ and $\Gamma(\cdot)$ is the Gamma function. ϵ is a fixed small constant.

The intrinsic reward is then given by:

$$r_t^i(o_t) = L(\xi) \left[\frac{\mu_k(\mathbf{e}_t, \xi)}{v_k(\mathbf{e}_t, \tilde{\xi}) + \epsilon} \right]^{1-\alpha} \quad (5)$$

Here, $L(\xi) = \tanh\left(\frac{1}{T} \sum_{t=0}^T \mu_1(\mathbf{e}_t, \xi)\right)$ is a scaling coefficient.

The choice of an encoder for intrinsic curiosity methods in DRL plays a pivotal role in representing the state for intrinsic reward computation, thus impacting the overall performance of the agent. We opt to use an IDM, which has demonstrated

superiority over other encoders, as shown by [27]. The model of the IDM is a neural network that predicts actions based on previous and current observations (Fig. 3c). The parameters of the IDM, denoted as ϕ , are optimized using a mean squared error (MSE) loss. The underlying intuition behind this encoder is that it captures features of the environment that are directly influenced by the robot's immediate controls. Thus, the exploration bonus received by the robot remains invariant to the environment's variability such as noises or changes in visual inputs.

D. Training protocol

Our agent is trained using the PPO algorithm [25]. Our choice is justified by the stability of this algorithm during training, its robustness to hyperparameters and its ability to parallelize multiple environments simultaneously. PPO is a policy gradient DRL algorithm that alternates between sampling data and optimizing a surrogate loss function.

During sampling, a set $\mathcal{M}$ of trajectories $\tau = (o_0, a_0, r_0, \dots, o_T, a_T, r_T)$ with length T is collected using the latest policy π_θ . The loss function of PPO is then given by:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{M}|T} \sum_{\tau \in \mathcal{M}} \sum_{t=0}^{T-1} \left[\min \left(R_t(\theta) \hat{A}_t, \mathcal{G}(R_t(\theta), \delta) \hat{A}_t \right) \right] \quad (6)$$

$\hat{A}_t$ is an estimate of the advantage function, given by:

$$A_t^{\pi_\theta}(o_t, a_t) = Q^{\pi_\theta}(a_t, o_t) - V^{\pi_\theta}(o_t) \quad (7)$$

The state-action value function is:

$$Q^{\pi_\theta}(a_t, o_t) = \mathbb{E}_{a_{t'} \sim \pi_\theta} \left[\sum_{t'=t}^T \gamma^{t'} r_{t'} \mid o_t, a_t \right] \quad (8)$$

The value function is given by

$$V^{\pi_\theta}(o_t) = \mathbb{E}_{a_{t'} \sim \pi_\theta} \left[\sum_{t'=t}^T \gamma^{t'} r_{t'} \mid o_t \right] \quad (9)$$

and $\mathcal{G}(\theta, \delta) = \text{clip}(R_t(\theta), 1 - \delta, 1 + \delta)$ is a clipping function with probability δ . $R_t(\theta) = \frac{\pi_\theta(a_t|o_t)}{\pi_{\theta_{old}}(a_t|o_t)}$ is the ratio of the probabilities of the current policy and the old policy $\pi_{\theta_{old}}$. We adopt a continuous actor-critic-style architecture with generalized advantage estimation (GAE) for an enhanced estimation of the value function [25].

IV. EXPERIMENTAL SETUP

In this section, we provide details on the implementation of our approach and describe our evaluation setup.

A. Environment

The entire training is conducted in simulation using Pybullet. We designed our environment to closely resemble our real scene. The scene was chosen to be an empty room with no objects and only a poster (Fig. 4). This decision was made for three main reasons: (i) it is easy to design such environments on simulators, (ii) to enable zero-shot transfer to the real robot by making the DRL model learn global

discriminative visual features, (iii) to prevent the robot from relying on visual features other than those from the desired image when it is not within the FOV of the camera. During training, the initial position of the robot is randomly sampled so that the initial distance error is in the range of 0.5m and 4.0m. The robot's initial orientation is randomly initialized to ensure that the desired image remains, even partially, within the camera's FOV.



Fig. 4: Images from (a) simulated and (b) real environments.

B. Training

Experiments were conducted using an NVIDIA RTX 3090 GPU and an Intel i9-12900KF CPU. We use the PPO agent with GAE provided by stable-baselines3 [28] to train our model. We follow the training protocol described in Sec. III-D to train our deep recurrent actor-critic policy in 48 parallel virtual environments, where each environment is initialized randomly. It takes around 2 hours to pass through one million training steps across all environments. The episode length T is set to 1500 steps.

In practice, we empirically set the parameters of our reward function in Eq. (2) λ_d and λ_β and K to 0.8, 0.2, and 0.5, respectively, to ensure a dense distribution of rewards in the training environment. The order of the divergence α is set to 0.5. The parameter η is a variable weighting parameter with an exponential decay set up similar as in [23].

C. Comparative study protocol

The following methods are compared in our experiments:

- **Random policy** is the simplest heuristic for the VS task. The robot selects random actions at each step.
- **IBVS** is a classical IBVS using point features from input images. To eliminate the influence of point extraction, tracking, and matching, we use the projection of the corners of the poster (Fig. 4a) as visual features.
- **LVS** is the VS approach proposed in [18]. This approach uses dense correspondence between the current and desired images as input for a DQN.
- **RPPO** is our base agent, a recurrent PPO without any intrinsic curiosity model.
- **RPPO+CNN (ours)** is our base agent combined with the intrinsic curiosity model introduced in Sec. III-C, and a fixed CNN encoder to represent observations.
- **RPPO+RND (ours)** is our base agent combined with the intrinsic curiosity method proposed in [21].
- **RPPO+IDM (ours)** is our base agent combined with the intrinsic curiosity model introduced in Sec. III-C, and an IDM as an encoder to embed observations.

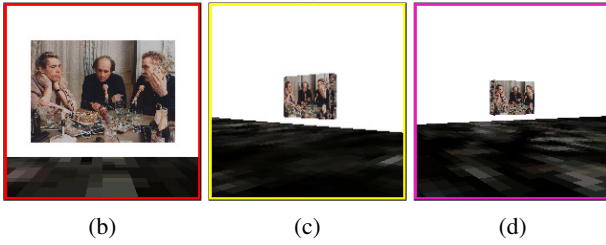
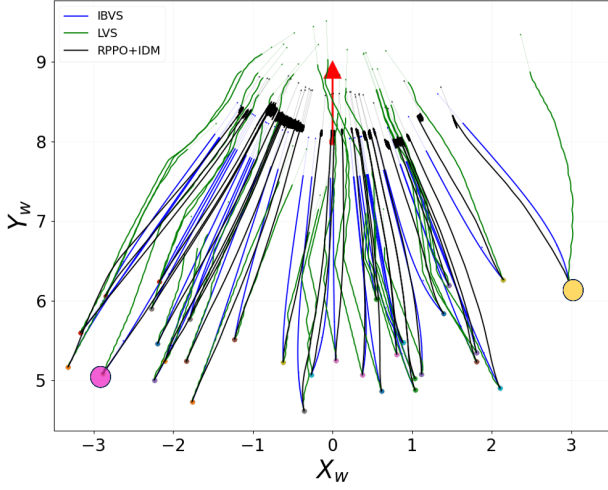


Fig. 5: Qualitative results: (a) Illustration of the performed trajectories by RPPO+IDM (black) compared to IBVS (blue) and LVS (green), (b) the target image acquired at the desired pose shown by a red arrow, and (c-d) examples of initial images acquired at the pose shown respectively by a yellow and a pink circle.

We use the error between the pose of the robot at convergence and the desired pose as a proxy for evaluating our VS task. For evaluation, we consider a fixed test set of 100 randomly pre-sampled initial poses. We report the performance in terms of position and orientation errors across our test set, calculated at convergence.

V. EXPERIMENTAL RESULTS

A. Simulation results

Fig. 5a shows the trajectories performed by our method (RPPO+IDM), IBVS and LVS. The target pose $\mathbf{p}^* = (0.0m, 8.0m, 0.0^\circ)$ is represented by the red arrow. The initial orientations and the orientations at convergence are represented with arrows for each corresponding method. The error along the X_w -axis is the highest for all the compared methods because of the nonholonomic constraints, which prevent the robot from displacing laterally. Nevertheless, our approach enables better convergence to the desired pose. More precisely, the performances of the different compared methods are assessed based on position errors ΔM and on orientation errors $\Delta\beta$ in Tab. I.

IBVS converges to a local optimum due to the coupled visual DOF, yielding a global optimum of the cost function in the feature space, despite the robot not reaching the

Method	$\Delta M(m)$	$\Delta\beta(^{\circ})$
random policy	2.9898	21.7380
IBVS	0.8779	17.3663
LVS	1.4420	18.8159
RPPO	0.6905	18.7987
RPPO+RND (ours)	0.6774	18.6153
RPPO+CNN (ours)	0.6442	17.5382
RPPO+IDM (ours)	0.5811	15.3208

TABLE I: The average pose errors at convergence across our test set. ΔM is the average position errors, $\Delta\beta$ is the average orientation errors.

desired pose. In contrast, LVS struggles to learn a good policy due to the discrete action space and the lack of temporal representation in the neural network model. RPPO performs better than the two previous methods; however, it exhibits limited performance due to the nature of exploration in DRL algorithms, which explore by sampling noise. This constitutes a major limitation to the application of DRL algorithms to the VS of nonholonomic robots. We postulate that the nonholonomic robot, restricted to two controllable DOF, faces the challenge of randomly sampling precise combinations of angular and linear velocities to displace laterally, thereby making learning complex trajectories very difficult.

Our methods achieve better results than all compared methods, namely IBVS, LVS, and RPPO (Tab. I). This result can be attributed to the incorporation of the LSTM layer, which captures temporal dependencies between observations, and the intrinsic curiosity model, which enables the robot to explore novel states, thereby discovering novel trajectories. As highlighted in Sec. III-C, the choice of an encoder for generating observation embeddings for the intrinsic curiosity model is crucial. RPPO+CNN, which employs a fixed random CNN encoder, yields better results than IBVS and LVS. However, the fixed random CNN encoder does not allow capturing features related to the task, resulting in underperformance compared to RPPO+IDM, which utilizes an IDM. We postulate that by learning sufficient features to represent the observation and using them as a means to calculate intrinsic rewards, the policy could utilize meaningful representations to overcome the variability caused by visual domain randomization.

Additionally, we combine the intrinsic curiosity model from [21] with our base agent RPPO (RPPO+RND) and compare it to the previous methods. RPPO+RND outperforms RPPO, LVS, and IBVS due to its ability to address the limitations present in these two methods. However, it falls short of our proposed method RPPO+IDM, as the intrinsic curiosity model in [21] lacks the capacity to capture features related to the task.

B. Reward functions comparison

We investigate the performance of our method under the reward function proposed in [18]:

$$r_t^e = \max\left(0, \min\left(\frac{d_{t-1}}{d_{init}}, 1\right) - \min\left(\frac{d_t}{d_{init}}, 1\right)\right) \quad (10)$$

with:

$d_t = f((x_t, y_t), (x^*, y^*)) + \lambda_\omega * f(\beta_t, \omega_t) + \lambda_\beta * f(\omega_t, \beta^*)$ where $f(x_1, x_2)$ is the euclidean distance between x_1 and x_2 . $\omega_t = \arctan(y^* - y_t, x^* - x_t)$ is an angle that is supposed to enforce the robot to rotate and align with the desired pose. As specified in [18], λ_ω is set to 0.2.

During a single episode, the agent aims to maximize its rewards. An optimal policy for our task should converge to a sum of rewards close to T . Under our reward function in Eq. (2), the agent can accumulate a high sum of rewards (Fig. 6), enabling efficient exploration of the environment. In contrast, we observe that the reward function in Eq. (10) impedes the agent's ability to learn the environment when used with our method. We postulate that this arises from the lack of mathematical coherence in the term d_t/d_{init} in Eq. (10), where the normalization of the reward term related to the angle ω_t is not consistent with respect to ω_{init} .

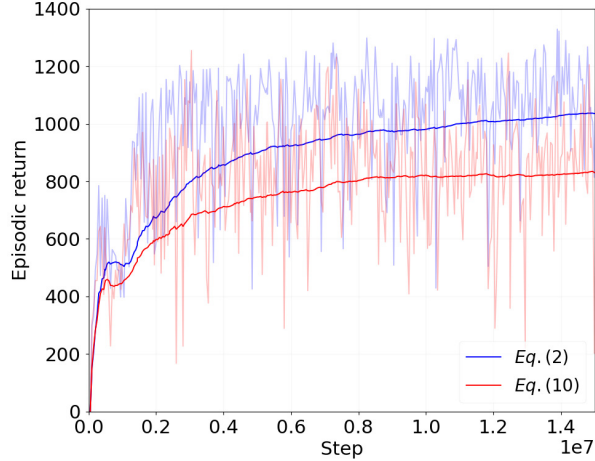


Fig. 6: Performance of the compared reward functions. The episodic return is the average of rewards perceived by the agent during an episode.

C. Real-world experiments

Real experiments are conducted with a four-wheeled nonholonomic mobile robot (Robotnik Summit-XL) with a calibrated perspective camera embedded frontward (Fig. 7).

Fig. 8 shows two experiments conducted on the real robot. In the first experiment, the desired image is fully visible within the camera FOV. The robot is approximately positioned at $2.0m$ from the desired pose with an angle of 15° that must be corrected. Our method demonstrates proficiency in handling these straightforward scenarios.



Fig. 7: Experimental real environment: a nonholonomic mobile robot (Robotnik Summit-XL differential base) with a front-mounted RGB camera. Note that the robotic arm is not used in this work.

In the second experiment, the desired image is barely visible in the camera FOV. The robot is approximately positioned at $2.5m$ from the desired pose with an angle of 40° that must be corrected. In such cases, IBVS fails due to the lack of useful visual features that could be used to compute the control velocities. In contrast, our DRL approach enables the robot to extract useful features and compute the appropriate velocities to reach the desired pose.

We emphasize that employing visual domain randomization is crucial for effectively transferring our control policy. Training the robot without randomization resulted in a poor performance in the real scene. The presence of noise in input images assists CNNs in learning robust features. At the same time, color randomization (Fig. 2) diversifies training data, allowing the agent to handle the changes in luminosity and contrast.

While trained only on synthetic data, without any particular effort to control the conditions of the real environment, our method transfers easily to the real robot. Our experiments show that our method transfers straightforwardly to the real robot and is capable of handling complex VS scenarios.

VI. CONCLUSION

We formulate the problem of learning visual servoing of nonholonomic mobile robots as a partially observable Markov decision process. We then propose a novel deep reinforcement learning method. Our approach incorporates a self-supervised intrinsic curiosity stage designed to bias the agent toward acquiring knowledge of complex trajectories. Our approach have the ability to learn useful features to reach the desired pose using reward signals, as opposed to classical methods, which often rely on minimizing errors in the image feature space. Moreover, we demonstrate the transferability of our learned model on a real robot.

However, our method has limitations, mainly due to the nature of deep reinforcement learning algorithms. Classical methods are plug-and-play and independent of specific environments, whereas deep reinforcement learning requires significant computational resources to train and generalize across different environments.

Future work includes using auxiliary tasks to learn task-related visual representations and training the model on a

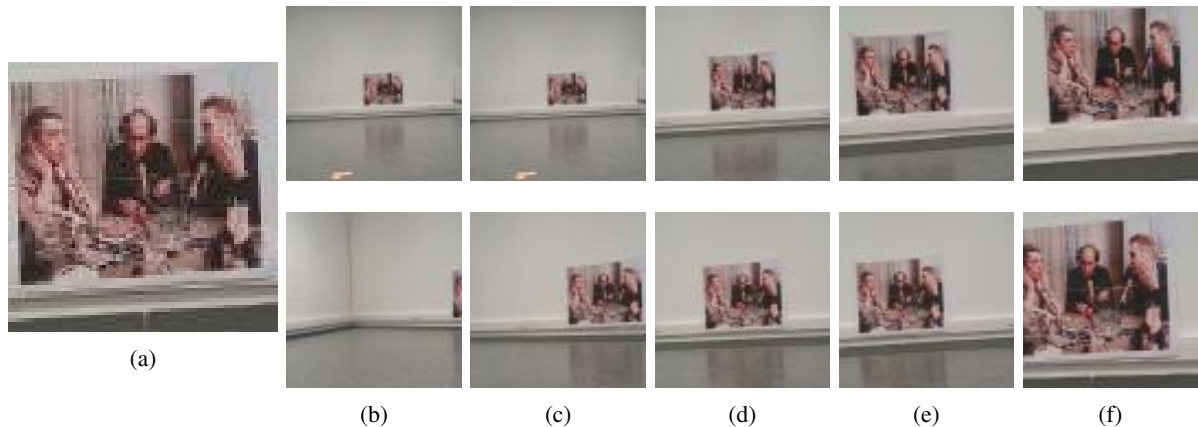


Fig. 8: Two experiments conducted on the real robot: experiments conducted on the real robot: (a) represents the desired image, (b) displays the initial image I_0 captured in the first experiment (top) and the second experiment (bottom), (c-e) are the transition images I_t acquired by the robot during the first experiment (top) and the second experiment (bottom), and (f) the final images I_T acquired at convergence.

large set of environments to study its generalization capabilities.

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